

Towards Robotic Trash Removal with Autonomous Surface Vehicles

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Abstract—Plastic pollution poses a critical threat to aquatic ecosystems, with floating debris such as bottles and bags disrupting water quality and endangering wildlife. This work presents a modular and simulation-validated framework for autonomous trash detection and collection using a vision-guided Autonomous Surface Vehicle (ASV). The system executes a boustrophedon (lawnmower-style) survey while enabling opportunistic detours, constrained to 25% of inter-line spacing, for intercepting visible debris. Crucially, the system prioritizes the integrity of the scientific survey, treating trash collection as a secondary objective to ensure the spatial fidelity of environmental measurements is not compromised. A lightweight color-segmentation proxy was integrated in a ROS–Gazebo simulation using the Heron USV, enabling real-time validation of perception-driven navigation logic. Visual triggers are filtered based on scan-zone gating to ensure spatially stable detour behavior. Trash is passively collected via a rear-mounted net mechanism. Across ten Monte Carlo trials, the system achieved an average coverage rate of 108.2%, a trash collection rate of 54%, and maintained 100% waypoint recovery, confirming robust multi-objective performance. Additionally, the trained YOLOv8n model was evaluated on live footage from the physical R/V Karin Valentine platform, validating detection reliability in real-world aquatic conditions. The overall architecture provides a scalable, resource-efficient solution for surface debris removal without distorting scientific data collection. Future work will focus on ROS 2 and PX4 integration and end-to-end deployment in urban water bodies such as Tempe Town Lake.

Index Terms—Visual servoing, Trash removal, Autonomous Surface Vehicle, Object tracking, autonomous navigation, environmental monitoring, marine robotics.

I. INTRODUCTION

Plastic pollution in aquatic environments continues to escalate, with recent estimates indicating that global plastic production exceeds 400 million tons annually, and millions of tons enter the ocean each year [18], [19]. A large portion of this waste predominantly single-use plastics such as bottles, wrappers, and bags accumulates in urban water bodies, threatening marine ecosystems and water quality. Traditional cleanup approaches, including manual collection and static

filtering systems, are labor-intensive, reactive, and often insufficient to address the dynamic and spatially distributed nature of surface-level debris. While robotic platforms and computer vision techniques have shown promise for environmental monitoring [3], [4], [20], few systems demonstrate real-time, adaptive trash interception that couples perception with autonomous control in constrained aquatic environments.

This paper presents a simulation-validated framework for real-time trash detection and interception using an autonomous surface vehicle. The system is composed of two main components: (1) a lightweight color-based detection module that emulates object recognition in a ROS-based simulation environment [24], and (2) a bounded detour control architecture that enables opportunistic collection of debris during a lawnmower survey mission. The system is deployed in a Gazebo Classic digital twin using the Heron Unmanned Surface Vehicle (USV) [7], modeling operations within a virtual representation of a water body with floating trash. Importantly, the system uses monocular vision only, avoiding the added hardware, power, and calibration overhead of stereo or depth sensors yet still achieving reliable spatial reasoning through scan-region gating and fixed-depth projection. Navigation logic respects a 25% lateral deviation constraint relative to the inter-line spacing, ensuring coverage continuity while allowing dynamic redirection in response to visual detections.

A key contribution of this work is the integration of a spatially constrained, perception-driven detour mechanism within a high-fidelity simulation environment. Floating trash objects are modeled as color-segmented targets and processed using a custom ROS node. Detour decisions are triggered only when detections fall within a defined vertical scan band, peripheral horizontal regions, and lie within 25% of the inter-line spacing from the boat’s path. This constraint-aware strategy ensures navigational stability while preserving the spatial integrity required for scientific surveys, such as temperature or turbidity mapping. By treating trash collection

as a secondary objective, the system prioritizes uninterrupted execution of the lawnmower survey. Across ten Monte Carlo trials, the framework achieved an average coverage rate of 108.08%, a trash collection success of 54%, and 100% way-point recovery following detours, demonstrating robust multi-objective performance without compromising the fidelity of primary environmental data collection.

In parallel, a trained YOLOv8 model was deployed and tested on the physical R/V Karin Valentine ASV using onboard camera footage. These field tests validated the model's reliability under real-world lighting and water conditions, forming a critical step toward transitioning the full perception-control pipeline to the hardware platform. This paper presents the validated simulation pipeline and field perception test as a foundational contribution toward autonomous aquatic trash removal. The modular framework supports scalable deployment in urban water bodies and establishes a clear pathway for future integration of the full control stack on embedded hardware.

II. RELATED WORK

Autonomous systems for trash detection and removal have been widely explored in recent years as plastic pollution poses growing environmental and logistical challenges. Existing research spans aerial, underwater, and surface-based robotic platforms equipped with visual perception systems for identifying and responding to marine debris.

Fulton et al. [12] evaluated four deep visual object detectors: YOLOv2, Tiny-YOLO, SSD, and Faster R-CNN for underwater trash detection. Their work demonstrated that deep learning models, when optimized through transfer learning, could operate on embedded devices and detect marine litter in real-time. Similarly, Kraft et al. [2] introduced the UAVVaste dataset for aerial trash detection and deployed a lightweight GPS-tagged vision system onboard an unmanned aerial vehicle (UAV), emphasizing the need for efficient, onboard inference in constrained settings. Maharjan et al. [11] extended this work to fluvial environments by comparing YOLOv5 and SSD for plastic detection in rivers using UAV sensor data. At the water surface, Suriya and Haldorai [10] proposed a vision-based trash collection robot using SSD. While effective for detection under structured conditions, their approach lacked dynamic navigation or real-time adaptation to new debris locations. Barrionuevo et al. [13] employed deep reinforcement learning to coordinate heterogeneous autonomous surface vehicles for plastic waste collection. Their work emphasized path optimization and cooperative task allocation, but did not incorporate direct visual perception for debris detection. In contrast to these approaches, our system integrates a real-time visual detection trigger with a constraint-aware navigation controller in a ROS-based Gazebo simulation environment using the Heron USV model. Unlike prior studies focused exclusively on UAV platforms or offline vision processing, our work simulates opportunistic trash interception using spatial deviation limits ($\leq 25\%$ of survey spacing) during an active coverage mission. Additionally, we validate the perception

module in the field by deploying a YOLOv8 model on the R/V Karin Valentine ASV, laying the groundwork for full integration of vision-driven navigation in real-world aquatic environments.

Outside the domain of trash collection, autonomous surface vehicles (ASVs) have been widely studied for their role in scientific environmental monitoring, particularly using structured coverage patterns. Sudha et al. [30] proposed a coverage planning algorithm for environmental monitoring using ASVs, showing boustrophedon patterns maximize uniform data density across survey lines. Karapetyan et al. [31] explored riverine surface vehicle navigation using lawnmower trajectories to collect bathymetric and environmental samples efficiently. Paull et al. [32] investigated sensor-driven online coverage planning with autonomous underwater vehicles, emphasizing adaptive boustrophedon strategies to accommodate terrain and data gradients. He et al. [33] applied boustrophedon decomposition to improve sampling performance in dynamic aquatic environments, demonstrating that structured path adherence preserves sampling resolution. These studies underscore the critical need for spatially uniform trajectory adherence in scientific missions, further motivating our constraint-aware opportunistic detour framework.

III. METHODOLOGY

This section presents the methodology adopted to design, simulate, and partially validate an Autonomous Surface Vehicle (ASV) for real-time floating trash detection and opportunistic collection in urban water environments. The overall system comprises three primary components: (i) a vision-based detection pipeline utilizing a YOLOv8n model fine-tuned for surface debris classification, (ii) a high-fidelity digital twin simulation developed in Gazebo Classic using the Heron USV and integrated with ROS Melodic, and (iii) a constraint-aware navigation controller that enables lateral detours within a 25% deviation envelope from a lawnmower survey path.

The ASV is designed to follow a boustrophedon (lawnmower-style) survey pattern as its primary coverage mission. Upon detecting floating trash, the boat performs a conditional detour based on spatial-filtering logic. Trash is only considered actionable if it (a) appears within the calibrated field of view and scanning range of the onboard camera, (b) lies within 25% of the distance between adjacent lines of the survey pattern, measured as the Euclidean distance between the boat and the detected object in local coordinates.

The system is developed and validated in simulation using the Heron USV model within the Gazebo Classic simulator, integrated with ROS Melodic [13]. Object detection, localization, and navigational decisions are continuously tested in a high-fidelity virtual lake setting before transitioning to hardware deployment. Rather than using complex manipulation systems, the platform is equipped with a rear-mounted passive net designed to collect debris as the boat approaches identified trash. This reduces mechanical complexity while enabling functional waste retrieval. As shown in Figure 1, the control architecture defines a conditional detour strategy

based on proximity and scan range constraints during the survey mission. This flowchart highlights the core contribution of our framework: integrating real-time debris detection and constrained navigation within a coverage mission, enabling opportunistic collection without compromising the fidelity of primary environmental data acquisition.

A. Trash Detection Pipeline

Recent studies have demonstrated the viability of using deep learning [25] for aquatic trash detection under real-time constraints. Fulton et al. [12] applied and evaluated four deep learning models: YOLOv2, Tiny-YOLO, SSD, and Faster R-CNN to detect marine debris in underwater scenes. Their study showed that with transfer learning and careful model configuration, real-time trash detection was feasible on resource-constrained embedded platforms. They also introduced a dataset featuring plastic debris, biological matter, and intentional man-made objects, and performed cross-device evaluations capturing mAP, IoU, and runtime performance. Their results suggested that deep visual detection systems can serve as the foundation for autonomous trash removal platforms, particularly in robotic underwater applications.

Similarly, Maharjan et al. [11] explored YOLOv5 and SSD for riverine plastic detection using UAV imagery, further validating the YOLO architecture’s suitability for environmental monitoring tasks. Other researchers, such as Suriya and Haldorai [8], developed SSD-based trash detection systems for surface-level garbage collection robots operating under structured conditions. Collectively, these works underscore the importance of optimizing detection models for real-time inference on embedded hardware.

Building on these insights, our work deploys a YOLOv8-based detection pipeline tailored for ASVs. YOLOv8 was selected for its architectural advancements over previous YOLO versions and its superior trade-off between accuracy and inference latency [4], [5]. The pipeline began with the YOLOv8n (nano) variant to ensure compatibility with embedded processors. Implemented in a Python 3.8 environment with Ultralytics’ YOLOv8 framework [4], the system integrates OpenCV and NumPy for preprocessing. The base model was tested on the COCO dataset for initial validation, then fine-tuned on the domain-specific *Waste in Water* dataset [7] to adapt it to real-world floating debris under varying visual conditions.

The *Waste in Water* dataset, sourced from RoboFlow [7], contains 3,798 images annotated across 19 classes of water-borne debris. The images were augmented with vertical and horizontal flips and resized to a consistent 640×640 pixels resolution. The dataset was pre-split into training, validation, and test sets (70%, 20%, 10%) and formatted to YOLOv8 standards. Model training was conducted for 50 epochs using the YOLOv8n backbone, achieving a validation mAP@0.5 of 0.264. The model demonstrated high detection reliability for certain classes, with “Plastic Bottle” reaching a class-wise mAP@0.5 of 0.878. A summary of detection performance is presented in Table I.

TABLE I: Validation Performance of YOLOv8n Model on Selected Trash Classes

Class	Images	Instances	Precision (P)	Recall (R)	mAP50	mAP50-95
All Classes	157	588	0.575	0.295	0.264	0.181
Plastic Bottle	31	31	0.902	0.871	0.878	0.752
Bottles	2	39	0.871	1.000	0.993	0.746
Plastic Bag	63	109	0.268	0.385	0.239	0.115
Can	5	10	1.000	0.000	0.000	0.000
Undefined Trash	18	49	0.167	0.082	0.066	0.045

During static image testing, up to 29 instances of plastic bottles were detected in a single frame with confidence values ranging from 0.314 to 0.938, as shown in Table II. Example outputs from the trained YOLOv8n model on aquatic images are shown in Figure 2, highlighting successful identification of floating plastic items such as bottles and bags.

TABLE II: Static image detection results using the YOLOv8n model

Image	Detected Classes	Instances	Confidence Range	Notes
Image 1	Plastic bottle	24	0.253–0.864	Strong focus on bottles
Image 2	Plastic Bottle, Plastic bag, Plastic bottles, Wood	8 (1, 1, 5, 1)	0.314–0.938	Varied classes, lower confidence for non-bottles

Video-based testing further confirmed the model’s robustness, with consistent detection of plastic bottles (50 instances, up to 0.82 confidence), frequent spotting of plastic cups (30 instances, 0.3–0.68), and episodic recognition of plastic bags (15 instances, peaking at 0.79), as detailed in Table III. Due to the dataset’s class imbalance where many trash types appeared infrequently a unified “trash” class strategy was proposed to improve generalization and simplify downstream navigation logic. This unification merges semantically similar labels such as “Plastic Bottle,” “Plastic Cup,” and “Plastic Bag” into a single detection category. Note: Some class labels (e.g., “Plastic Bottle” vs. “Plastic bottle”) reflect minor inconsistencies in dataset annotation.

The final model was integrated with both static image and video feed pipelines, enabling real-time trash detection with CPU inference times averaging 25–35 milliseconds per frame. Figure 3 illustrates temporal consistency in trash detection from a sample video feed, with plastic debris identified across multiple frames. This perception pipeline forms the backbone of the ASV, providing real-time input to the autonomous navigation module for on-the-fly decision-making during trash encounters.

TABLE III: Video-based detection results using the YOLOv8n model

Class	Instances	Confidence Range	Notes
Plastic bottle	50	0.27–0.82	Most consistent, peaks late
Plastic cup	30	0.25–0.68	Frequent but lower confidence
Plastic bag	15	0.25–0.79	Sporadic, stronger later
Plastic Bottle	10	0.27–0.56	Possible overlap with lowercase

B. Digital Twin and Simulation Environment

Simulation-based validation is a widely adopted method in robotics for verifying perception-control pipelines before physical deployment. ROS-integrated simulators such as Gazebo offer modular and high-fidelity environments for

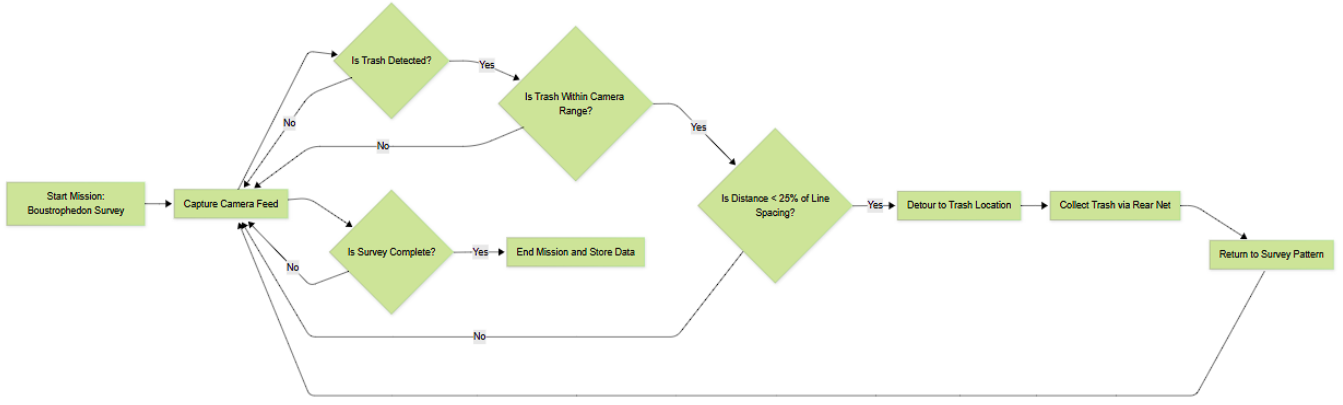


Fig. 1: Visual Perception-Based Conditional Detour Logic for Boustrophedon Survey Navigation



Fig. 2: Testing detection of trash on test images



Fig. 3: Detection of trash on a test video

testing robotic systems under configurable and repeatable conditions. Prior work has explored the use of ROS and Gazebo for aquatic robots targeting waste removal. SaiTeja and Pramod [14] developed a ROS-based simulation of an autonomous surface vehicle (ASV) for trash interception. Similarly, Sanfilippo et al. [15] built a digital twin of the Heron Unmanned Surface Vehicle (USV) using ROS and Gazebo to support lightweight experimentation on water-based robotic platforms. Lindsay et al. [16] used the Clearpath Heron model to simulate multi-domain sensing scenarios with collaborative marine robots, and Olsen [17] designed a ROS-based USV path planner specifically for trash collection applications.

Building on these foundations, our simulation framework uses the Gazebo Classic simulator integrated with ROS Melodic [9]. The digital twin replicates the Heron USV’s operational behavior in a virtual environment approximating the conditions water body with floating trash objects. This setup allows end-to-end validation of the real-time perception

and navigation logic with full control over sensor input, environmental conditions, and trash placement.

1) *System Design and Configuration:* The simulation environment was implemented using ROS Melodic integrated with the Gazebo Classic simulator. A digital twin of the Heron Unmanned Surface Vehicle (USV) was adapted to include an RGB camera mounted at the bow with a 90° field of view and 640×480 pixel resolution. This configuration provided synthetic image streams for perception testing within a controlled aquatic environment.

Floating debris was represented using buoyant, collidable 3D models placed at randomized surface locations. To emulate real-time object detection without the computational overhead of running a full neural network, a lightweight color segmentation node was developed using ROS-compatible image processing tools. This proxy served as a stand-in for the actual YOLOv8-based perception module and enabled efficient validation of the integrated control logic. The overall setup allowed systematic evaluation of detection-triggered navigation within a closed-loop simulation. A representative snapshot of the simulation environment is shown in Figure 4.

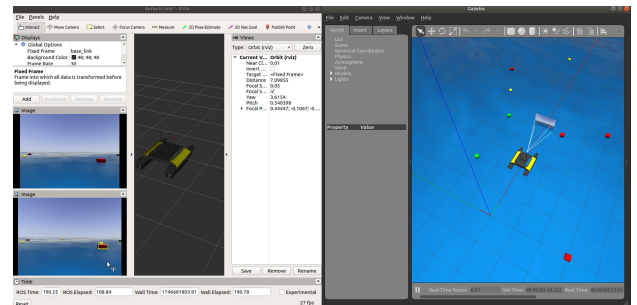


Fig. 4: Digital Twin Simulation Environment with Heron USV

2) *Visual Trigger Logic and Scan-Zone Filtering:* To ensure stable and goal-consistent navigation, the ASV’s detour decisions depend not only on the presence of a detected object but also on its spatial position within the camera’s field of view. Because monocular cameras lack depth perception, a fixed-

depth assumption is used to approximate relative position. A spatial scan-zone filtering strategy is implemented to gate detections based on geometric relevance.

Specifically, trash detection is prioritized only when the bounding box centroid lies within:

- A vertical band spanning 10%–40% of the image height, to exclude objects too close or far from the vehicle.
- Peripheral horizontal bands, either left of 34% or right of 66% of the image width, to enforce clear directional detours and avoid indecisive mid-field triggers.

Detections falling outside these zones especially in the central 34%–66% horizontal band or extreme top/bottom areas are ignored to maintain navigational stability and avoid erratic movements.

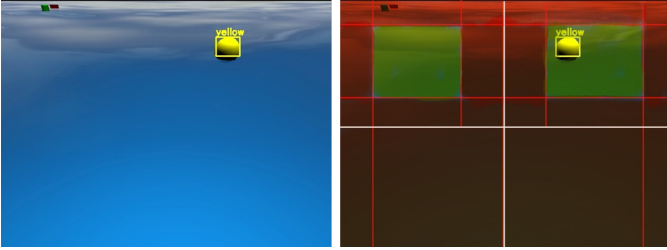


Fig. 5: Left: Raw camera feed showing a detected trash object. Right: Zone-based filtering overlay; only detections within the central vertical band (green) and horizontal periphery (red zones) are considered for detour triggering. Midfield detections are ignored to ensure robust trajectory control.

Figure 5 illustrates the spatial filtering logic. Custom ROS nodes process the image stream, evaluate bounding box centroids, and apply zone-based gating to determine detour eligibility. Vertical and horizontal constraints balance responsiveness with stability in opportunistic trash interception.

3) *Simulation Setup and Navigation Logic:* The USV performs a boustrophedon (lawnmower) survey pattern across a 12 m $\times$ 14 m virtual lake area. Navigation commands are handled by a mission planner node that integrates YOLO-based detections with a path deviation logic. Detected objects are filtered based on: (i) confidence $\geq 40\%$, (ii) presence within the camera’s central scan zone, and (iii) distance $\leq 25\%$ of inter-line spacing.

TABLE IV: Simulation Environment Configuration

Component	Configuration
Simulator	Gazebo Classic 11
ROS Version	ROS Melodic on Ubuntu 18.04
Robot Model	Heron USV (Clearpath Robotics)
Camera Plugin	gazebo_ros_camera, 90° FOV, 640 $\times$ 480 px
Vision Pipeline	Color Detection via ROS topic integration /front_camera/image_raw
Trash Models	URDFs with buoyancy + collision
Survey Pattern	Boustrophedon (lawnmower), 25% detour rule
Trials	10 randomized Monte Carlo runs

Upon satisfying these criteria, the controller initiates a lateral detour toward the detected trash. If the object enters the simulated interception zone (at the rear of the boat), it is marked as collected. The vehicle then realigns to the nearest waypoint and resumes its survey trajectory.

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4) *Performance Testing and Evaluation:* To evaluate performance, ten randomized Monte Carlo trials were conducted. Each trial introduced trash objects at varied positions, orientations, and lighting conditions. Key metrics recorded included: number of trash detections, detour activations, collection counts, and trajectory deviation. These logs were processed into CSV format and visualized for qualitative and quantitative analysis.

IV. RESULTS AND EVALUATION

This section presents the results of simulation experiments conducted to validate the integrated perception and navigation framework under mission-relevant constraints. The ASV was evaluated in a 14 m $\times$ 12 m virtual lake environment designed in Gazebo Classic. The platform followed a boustrophedon survey pattern while opportunistically detouring to collect floating trash detected via a color-segmentation proxy. The objective was to assess whether collection could be achieved without compromising area coverage or navigational stability. The trials followed the visual and spatial constraints outlined in Section III, including scan-zone filtering and lateral deviation limits to 25% of inter-line spacing.

A. Quantitative Results

The system consistently maintained high mission fidelity across trials. Table V summarizes average metrics recorded across ten runs. The system achieved an average coverage of 108.47%, resulting from minor overshoots during semicircular turns and detour recovery, while maintaining full adherence to the planned survey path. Trash collection averaged 52% across all trials, with a maximum of 70% in individual runs. All detours respected the spatial deviation constraint and rejoined the path without manual intervention.

TABLE V: Summary of Simulation Performance Across 10 Trials

Metric	Average Value
Coverage Achieved	108.38%
Trash Collection Rate	52.0%
Detour Activations per Trial	3.4
Avg. Path Deviation (m)	0.34 m
Deviation as % of Inter-line Spacing	11.3%
Waypoint Recovery Success	100%

B. Trajectory Evaluation

Figure 6 illustrates a representative simulation trial with overlaid way-points, actual vehicle trajectory, and trash locations. The robot smoothly deviated to intercept valid targets and consistently returned to its original path, confirming both controller stability and detour logic reliability.

These results validate the efficacy of the proposed system for autonomous trash interception without compromising coverage goals. The use of monocular vision, spatial filtering, and deviation-constrained control enables effective opportunistic trash collection while maintaining high survey fidelity. Detour

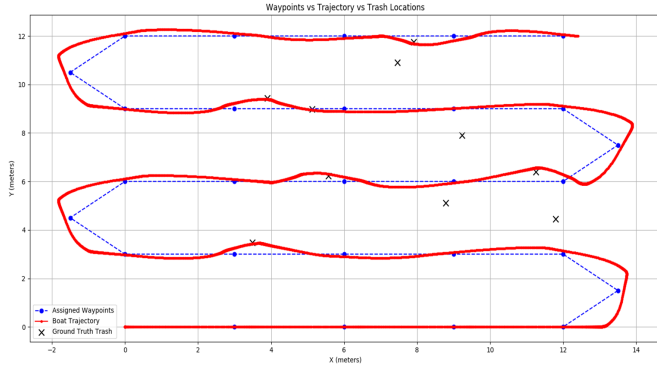


Fig. 6: Representative simulation trajectory showing the planned boustrophedon path (blue dashed), actual trajectory (red), and trash positions (black crosses). The system demonstrates bounded detours and reliable path recovery.

activations remained bounded, and all trials achieved full way-point recovery, confirming robustness of the navigation pipeline.

C. Field Survey Structure

Figures 7 and 8 illustrate the type of scientific data collected during environmental surveys conducted with the R/V Karin Valentine ASV. Figure 7 shows a bathymetric depth field mapped using a structured lawnmower (boustrophedon) trajectory, highlighting the importance of consistent spatial coverage in environmental sampling. Complementing this, Figure 8 presents collected data analysis from the same platform on December 6, visualizing the Gaussian process mean (top row) and uncertainty (bottom row) maps generated using stationary kernels. These examples underscore the need to preserve survey geometry and spatial sampling fidelity when integrating real-time opportunistic objectives such as trash interception. While our current trash collection system is simulation-based, future work will focus on field integration with bounded detours that do not compromise the integrity of environmental measurements such as bathymetry, temperature, or turbidity.

V. CONCLUSION

This work presents a modular and simulation-validated framework for autonomous surface-level trash removal, leveraging a real-time perception-control loop onboard an autonomous surface vehicle (ASV). The system integrates a vision-based detection module with a constraint-aware navigation controller to enable opportunistic trash interception without compromising coverage. A passive collection mechanism eliminates the need for complex actuators, while a bounded detour strategy ensures adherence to the primary survey objective. The digital twin system, developed using the Heron USV in ROS-Gazebo, enabled closed-loop testing of detection, control, and recovery behavior. Detour triggers were activated only when floating debris met strict spatial criteria appearing within a designated scan zone and no more than 25% laterally offset from the survey line.

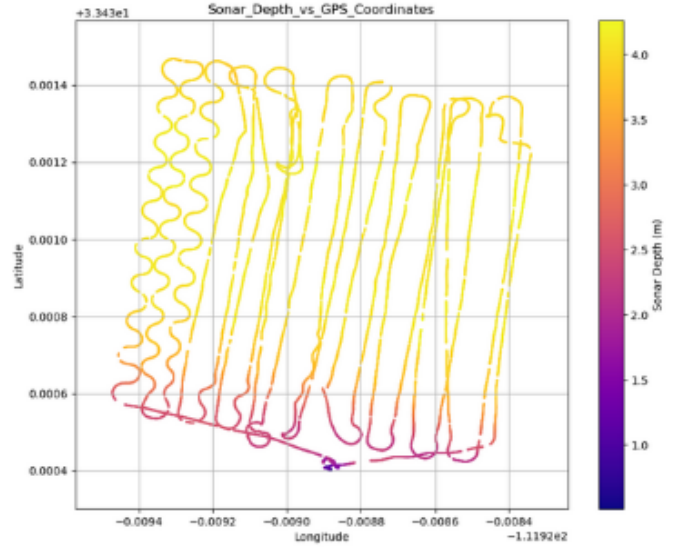


Fig. 7: Bathymetric Depth plotted against GPS from the RV Karin Valentine ASV trial overlaid with lawnmower trajectory.

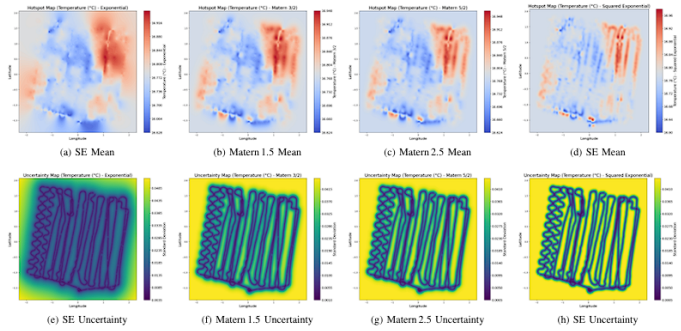


Fig. 8: Collected data analysis (Dec. 6): mean (top row) and uncertainty (bottom row) maps for stationary kernels

Across ten Monte Carlo trials, the system achieved an average coverage rate of 108.2%, consistently exceeding baseline survey expectations. The average trash collection rate was 54%, with a maximum of 70% in certain scenarios. Lateral deviation remained within 11.4% of inter-line spacing on average, and 100% waypoint recovery was observed following all detours, confirming the controller's stability and robustness. Field validation was also conducted on the R/V Karin Valentine, where the YOLOv8-based detection module reliably identified plastic debris in real-time using monocular RGB input under natural lighting. These results confirm the feasibility of integrating opportunistic trash removal within routine coverage missions.

Overall, the proposed framework offers a scalable and resource-efficient solution for real-world surface debris collection. Future work will focus on full system integration with PX4 [21] and ROS 2, embedded deployment, and live field trials in urban water bodies such as Tempe Town Lake.

As this system advances toward real-world deployment, a

key consideration will be ensuring that opportunistic navigation for trash collection does not compromise the spatial integrity of scientific surveys. Field missions conducted on the R/V Karin Valentine demonstrate the importance of maintaining structured sampling patterns—such as boustrophedon coverage for accurate collection of spatially dependent data like bathymetry, turbidity, or temperature fields. Preserving this fidelity while enabling minimal, bounded detours for debris interception will be essential to integrating trash removal capabilities into ongoing ecological monitoring workflows.

VI. FUTURE WORK

While the current simulation-based framework demonstrates stable and constraint-aware trash interception behavior, several components remain to be integrated for full deployment on a physical ASV. The simulation trials employed a simplified perception proxy using color-based segmentation, enabling robust validation of detour logic and trajectory recovery. However, transitioning to real-world operation introduces new challenges related to sensor fidelity, environmental variability, and embedded system integration. The immediate next step involves deploying the full YOLOv8n detection pipeline onboard the R/V Karin Valentine platform. This includes real-time RGB data acquisition, model inference, spatial filtering, and trajectory modulation, all handled via a ROS 2-based architecture. Preliminary field tests with live onboard camera input have confirmed the model’s ability to detect floating debris under natural lighting and water reflection conditions.

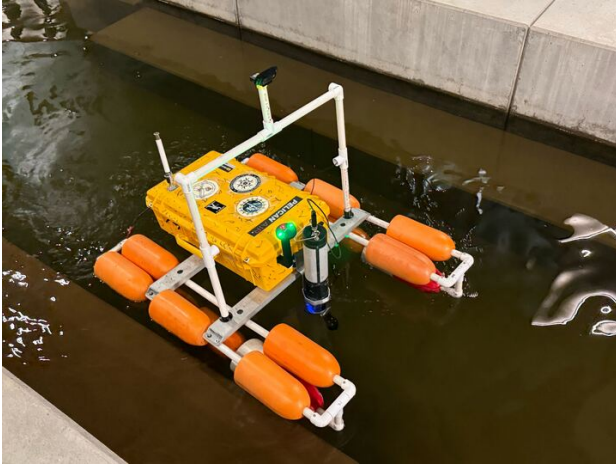


Fig. 9: Camera Installation on R/V Karin Valentine

Ongoing work includes integration of the autonomous navigation stack, respecting the same spatial constraints validated in simulation namely a lateral detour cap of 25% inter-line spacing, selective scan-band filtering, and waypoint recovery. Field deployment will evaluate system robustness under challenging visual and dynamic aquatic conditions, such as surface turbulence, occlusion, and lighting transitions.

Real-time telemetry, including detection logs, deviation metrics, and sensor fusion data, will be recorded using ROS 2’s native logging framework for post-mission evaluation. The

control stack will be migrated to a PX4-compatible setup, interfacing the Odroid-M1 onboard computer) [23] with the Hex Cube Black flight controller) [22] via MAVROS for hardware-level autonomy.

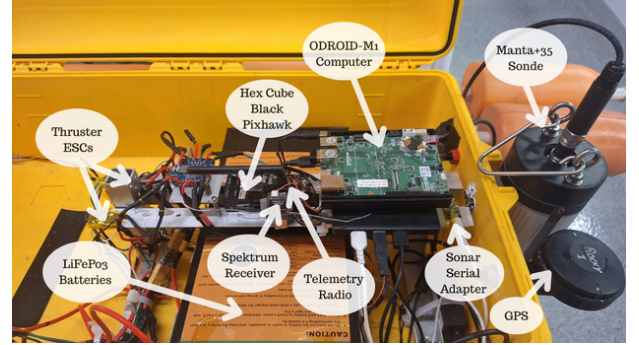


Fig. 10: R/V Karin Valentine Hardware Stack Integration

The vessel will also integrate a suite of environmental sensors for future multi-modal missions, [including the Blue Robotics 1D Ping Echosounder [28] for bathymetry and Eureka Manta+35 [29] for biogeochemical profiling. ROS 2 nodes will manage sensor synchronization and timestamped logging for mission analytics. In summary, upcoming development will focus on real-world validation of the full perception-control pipeline, robust controller deployment using PX4) [21] + ROS2 Jazzy) [27], and autonomous surface debris interception in live aquatic environments such as Tempe Town Lake.



Fig. 11: Real-World Detection Output: YOLOv8n Inference on R/V Karin Valentine ASV

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